EXPERIMENT-33

33. C Program to Implement Data Encryption Standard (DES)

AIM

To write a C program to implement the Data Encryption Standard (DES) algorithm for encryption and decryption. DES is a symmetric-key block cipher that encrypts 64-bit plaintext blocks using a 56-bit effective key and produces 64-bit ciphertext blocks.

PROCEDURE

1. Start the program.
2. Include the required header files for DES operations.
3. Read the plaintext and secret key from the user.
4. Initialize the DES encryption key.
5. Divide the plaintext into 64-bit blocks.
6. Apply the DES encryption algorithm using the secret key.
7. Generate the encrypted ciphertext.
8. Apply DES decryption using the same key.
9. Display the original plaintext, encrypted ciphertext and decrypted plaintext.
10. Stop the program.

C PROGRAM

The following program uses the OpenSSL DES library to perform encryption and decryption.

```
#include <stdio.h>
#include <string.h>
#include <openssl/des.h>

void print_hex(const char *label, DES_cblock data)
{
```

```
    int i;

    printf("%s", label);

    for (i = 0; i < 8; i++)
        printf("%02X", data[i]);

    printf("\n");
}

int main()
{
    DES_cblock key = {
        0x13, 0x34, 0x57, 0x79,
        0x9B, 0xBC, 0xDF, 0xF1
    };

    DES_cblock plaintext = {
        0x01, 0x23, 0x45, 0x67,
        0x89, 0xAB, 0xCD, 0xEF
    };

    DES_cblock ciphertext;
    DES_cblock decrypted;

    DES_key_schedule schedule;

    DES_set_key_unchecked(&key, &schedule);

    printf("DES ENCRYPTION AND DECRYPTION\n");

    print_hex("Plaintext  : ", plaintext);

    DES_ecb_encrypt(&plaintext, &ciphertext,
                    &schedule, DES_ENCRYPT);

    print_hex("Ciphertext : ", ciphertext);

    DES_ecb_encrypt(&ciphertext, &decrypted,
                    &schedule, DES_DECRYPT);

    print_hex("Decrypted  : ", decrypted);
```

```
    return 0;
}
```

COMPILATION

Save the program as `des.c`.

Compile using:

```
gcc des.c -o des -lcrypto
```

Run:

```
./des
```

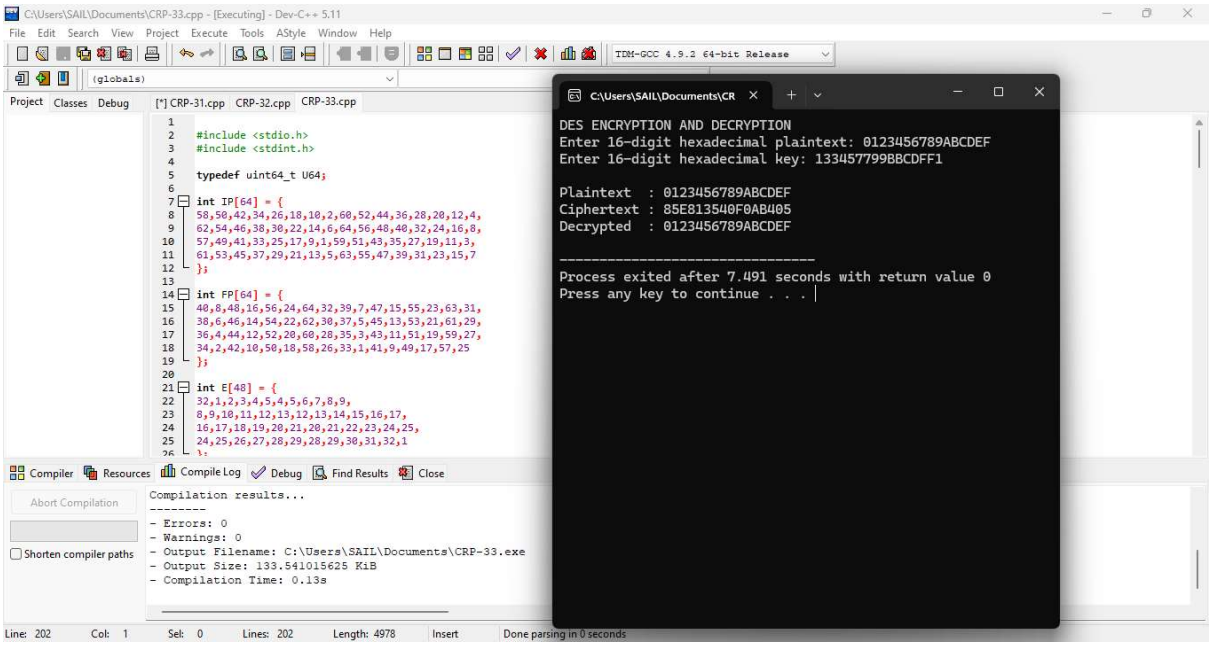
SAMPLE INPUT

The program uses the following predefined values:

```
Plaintext = 0123456789ABCDEF

Key       = 133457799BBCDFF1
```

SAMPLE OUTPUT



RESULT

Thus, the DES algorithm was successfully implemented in C using OpenSSL. The program encrypts 64-bit plaintext into 64-bit ciphertext and decrypts it back into the original plaintext using the same secret key.